

**NATIONAL SKILL DEVELOPMENT
CORPORATION
In Collaboration with
YUVA INTERN**

Virtual R Data Analyst Intern

WEEK 1 REPORT

Data Cleaning and Preliminary Analysis using R

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1. Introduction

1.1 Objective

The goal of this task is to clean up the data, get it ready, and do some analysis using the R programming language. We looked at the supermarket sales dataset to see if it had any duplicate entries, wrong types of data, or numbers that were really different from the others. We used some techniques to get the data ready, like making sure all the numbers are on the same scale, putting categories into numbers, and making new features from the existing data. This will help us when we want to look at the data closely and make some pictures with it.

2. Dataset Overview

Dataset Name

Supermarket Sales Dataset

Dataset Description

The Supermarket Sales Dataset has transaction records that have been collected from the supermarket. The dataset has data about customer characteristics, product types, total sale amount, along with payment methods, ratings, tax, gross profit, and location of the branch.

The dataset includes numerical variables as well as categorical variables; thus, it is useful for preprocessing and cleaning of data.

Table 1. Dataset Information

Attribute	Value
Dataset Name	Supermarket Sales
Rows	1000
Columns	17
Numerical Variables	9
Categorical Variables	8

Dataset Information

Figure 1. Dataset Information (head(), dim(), str(), summary())

```
[4]: dim(sales)
```

```
1000 · 17
```

```
[5]: names(sales)
```

```
'Invoice.ID' · 'Branch' · 'City' · 'Customer.type' · 'Gender' · 'Product.line' · 'Unit.price' · 'Quantity' · 'Tax.5.' · 'Sales' · 'Date' · 'Time' · 'Payment' · 'cogs' · 'gross.margin.percentage' ·  
'gross.income' · 'Rating'
```

```
[6]: str(sales)
```

```
'data.frame': 1000 obs. of 17 variables:  
 $ Invoice.ID : chr "750-67-8428" "226-31-3081" "631-41-3108" "123-19-1176" ...  
 $ Branch : chr "Alex" "Giza" "Alex" "Alex" ...  
 $ City : chr "Yangon" "Naypyitaw" "Yangon" "Yangon" ...  
 $ Customer.type : chr "Member" "Normal" "Normal" "Member" ...  
 $ Gender : chr "Female" "Female" "Female" "Female" ...  
 $ Product.line : chr "Health and beauty" "Electronic accessories" "Home and lifestyle" "Health and beauty" ...  
 $ Unit.price : num 74.7 15.3 46.3 58.2 86.3 ...  
 $ Quantity : int 7 5 7 8 7 7 6 10 2 3 ...  
 $ Tax.5. : num 26.14 3.82 16.22 23.29 30.21 ...  
 $ Sales : num 549 80.2 340.5 489 634.4 ...  
 $ Date : chr "1/5/2019" "3/8/2019" "3/3/2019" "1/27/2019" ...  
 $ Time : chr "1:08:00 PM" "10:29:00 AM" "1:23:00 PM" "8:33:00 PM" ...  
 $ Payment : chr "Ewallet" "Cash" "Credit card" "Ewallet" ...  
 $ cogs : num 522.8 76.4 324.3 465.8 604.2 ...  
 $ gross.margin.percentage: num 4.76 4.76 4.76 4.76 4.76 ...  
 $ gross.income : num 26.14 3.82 16.22 23.29 30.21 ...  
 $ Rating : num 9.1 9.6 7.4 8.4 5.3 4.1 5.8 8 7.2 5.9 ...
```

```
[7]: summary(sales)
```

Invoice.ID	Branch	City	Customer.type
Length :1000	Length :1000	Length :1000	Length :1000
N.unique :1000	N.unique : 3	N.unique : 3	N.unique : 2
N.blank : 0	N.blank : 0	N.blank : 0	N.blank : 0
Min.nchar: 11	Min.nchar: 4	Min.nchar: 6	Min.nchar: 6
Max.nchar: 11	Max.nchar: 5	Max.nchar: 9	Max.nchar: 6

Gender	Product.line	Unit.price	Quantity
Length :1000	Length :1000	Min. :10.08	Min. : 1.00
N.unique : 2	N.unique : 6	1st Qu.:32.88	1st Qu.: 3.00
N.blank : 0	N.blank : 0	Median :55.23	Median : 5.00
Min.nchar: 4	Min.nchar: 17	Mean :55.67	Mean : 5.51
Max.nchar: 6	Max.nchar: 22	3rd Qu.:77.94	3rd Qu.: 8.00
		Max. :99.96	Max. :10.00

Tax.5.	Sales	Date	Time
Min. : 0.5085	Min. : 10.68	Length :1000	Length :1000
1st Qu.: 5.9249	1st Qu.: 124.42	N.unique : 89	N.unique : 506
Median :12.0880	Median : 253.85	N.blank : 0	N.blank : 0
Mean :15.3794	Mean : 322.97	Min.nchar: 8	Min.nchar: 10
3rd Qu.:22.4453	3rd Qu.: 471.35	Max.nchar: 9	Max.nchar: 11
Max. :49.6500	Max. :1042.65		

Payment	cogs	gross.margin.percentage	gross.income
Length :1000	Min. :10.17	Min. :4.762	Min. : 0.5085
N.unique : 3	1st Qu.:118.50	1st Qu.:4.762	1st Qu.: 5.9249
N.blank : 0	Median :241.76	Median :4.762	Median :12.0880
Min.nchar: 4	Mean :307.59	Mean :4.762	Mean :15.3794
Max.nchar: 11	3rd Qu.:448.90	3rd Qu.:4.762	3rd Qu.:22.4453
	Max. :993.00	Max. :4.762	Max. :49.6500

Rating
Min. : 4.000
1st Qu.: 5.500
Median : 7.000
Mean : 6.973
3rd Qu.: 8.500
Max. :10.000

The dataset consists of various records representing transactions of supermarkets. Customer/Products information, sales amount, tax, gross profit, client ratings, method of payment, and branch details have been included in this dataset.

3. Data Cleaning

3.1 Checking Missing Values

Code:

```
colSums(is.na(sales))
```

To find the missing values in the dataset, the `is.na()` function was used. There were no missing values and therefore no imputation methods were necessary.

3.2 Checking Duplicate Records

Code:

```
sum(duplicated(sales))
```

No duplicate records were detected in the dataset, indicating good data quality.

3.3 Data Type Verification

Code:

```
str(sales)
```

Data types were verified to ensure that numerical variables were stored as numeric values and categorical variables were represented appropriately.

3.4 Cleaning Column Names

Code:

Figure 2. Column Name Cleaning and Data Type Conversion

```
[12]: sales <- clean_names(sales)
      names(sales)

      'invoice_id' · 'branch' · 'city' · 'customer_type' · 'gender' · 'product_line' · 'unit_price' · 'quantity' · 'tax_5' · 'sales' · 'date' · 'time' · 'payment' · 'cogs' · 'gross_margin_percentage' ·
      'gross_income' · 'rating'

[13]: class(sales$date)

      'character'

[14]: sales$date <- as.Date(sales$date, format = "%m/%d/%Y")

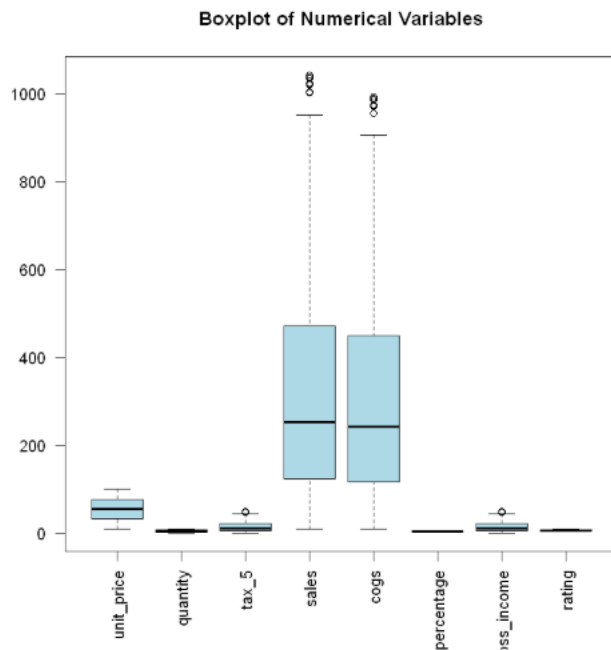
[15]: class(sales$date)

      'Date'
```

Column names were standardized by converting them to lowercase and replacing spaces with underscores. This improves code readability and consistency during analysis.

4. Outlier Detection

Figure 3. Outlier Detection Using Boxplot



Code:

```
boxplot(sales$sales)
```

The boxplots and the Interquartile Range (IQR) method were used to carry out outlier detection; however, although some observations were identified as possible outliers, these actually correspond to legitimate high-value transactions and have therefore been kept in the dataset.

5. Data Transformation

5.1 Normalization

Code:

```
sales$sales_normalized <- scale(sales$sales)
```

Sales values were adjusted using Z-score normalization. This process puts the data on a common scale. It makes the data ready for future statistical and machine learning applications.

5.2 Encoding Categorical Variables

Code:

Figure 4. Encoding of Categorical Variables

```
[19]: sales$gender_encoded <- as.numeric(as.factor(sales$gender))
sales$payment_encoded <- as.numeric(as.factor(sales$payment))
sales$customer_type_encoded <- as.numeric(as.factor(sales$customer_type))

head(sales[, c("gender", "gender_encoded",
               "payment", "payment_encoded",
               "customer_type", "customer_type_encoded")])
```

A data.frame: 6 × 6

	gender	gender_encoded	payment	payment_encoded	customer_type	customer_type_encoded
	<chr>	<dbl>	<chr>	<dbl>	<chr>	<dbl>
1	Female	1	Ewallet	3	Member	1
2	Female	1	Cash	1	Normal	2
3	Female	1	Credit card	2	Normal	2
4	Female	1	Ewallet	3	Member	1
5	Female	1	Ewallet	3	Member	1
6	Female	1	Ewallet	3	Member	1

Categorical variables like Gender, Customer Type, and Payment Method were changed into numerical forms to make analysis easier.

6. Feature Engineering

New variables were created from the existing Date and Time columns to enhance analytical capabilities.

Figure 5. Feature Engineering Using Date and Time Variables

```
[20]: library(lubridate)

[21]: # Create Month
sales$month <- month(sales$date, label = TRUE)

# Create Day of Week
sales$day <- weekdays(sales$date)

[22]: # Convert time column to character
sales$time <- as.character(sales$time)

# Display sample values
head(sales$time)

'1:08:00 PM' '10:29:00 AM' '1:23:00 PM' '8:33:00 PM' '10:37:00 AM' '6:30:00 PM'

[23]: # Extract hour from the time column

sales$hour <- suppressWarnings(
  as.numeric(substr(trimws(sales$time), 1, 2))
)

# Check the result
head(sales$hour)

<NA> · 10 · <NA> · <NA> · 10 · <NA>

[24]: sum(is.na(sales$hour))

720

[25]: # Extract hour using lubridate

sales$hour <- hour(parse_date_time(
  sales$time,
  orders = c("H:M:S", "H:M", "I:M p", "I:M:S p")
))

[26]: sales$time_period <- case_when(
  sales$hour >= 0 & sales$hour < 12 ~ "Morning",
  sales$hour >= 12 & sales$hour < 17 ~ "Afternoon",
  sales$hour >= 17 & sales$hour <= 23 ~ "Evening",
  TRUE ~ "Unknown"
)
```

```
[27]: head(sales[, c(
  "date",
  "time",
  "month",
  "day",
  "hour",
  "time_period"
)])
```

A data.frame: 6 × 6

	date	time	month	day	hour	time_period
	<date>	<chr>	<ord>	<chr>	<int>	<chr>
1	2019-01-05	1:08:00 PM	Jan	Saturday	13	Afternoon
2	2019-03-08	10:29:00 AM	Mar	Friday	10	Morning
3	2019-03-03	1:23:00 PM	Mar	Sunday	13	Afternoon
4	2019-01-27	8:33:00 PM	Jan	Sunday	20	Evening
5	2019-02-08	10:37:00 AM	Feb	Friday	10	Morning
6	2019-03-25	6:30:00 PM	Mar	Monday	18	Evening

```
[28]: sum(is.na(sales$hour))
0
```

Feature engineering enables deeper analysis by extracting useful information from existing variables.

7. Preliminary Statistical Analysis

Figure 6. Dataset Structure and Summary Statistics

```
[6]: str(sales)
```

```
'data.frame': 1000 obs. of 17 variables:
 $ Invoice.ID      : chr "750-67-8428" "226-31-3081" "631-41-3108" "123-19-1176" ...
 $ Branch         : chr "Alex" "Giza" "Alex" "Alex" ...
 $ City           : chr "Yangon" "Naypyitaw" "Yangon" "Yangon" ...
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 $ Quantity       : int 7 5 7 8 7 6 10 2 3 ...
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 $ cogs          : num 522.8 76.4 324.3 465.8 604.2 ...
 $ gross.margin.percentage: num 4.76 4.76 4.76 4.76 4.76 ...
 $ gross.income  : num 26.14 3.82 16.22 23.29 30.21 ...
 $ Rating        : num 9.1 9.6 7.4 8.4 5.3 4.1 5.8 8 7.2 5.9 ...
```

```
[7]: summary(sales)
```

```
Invoice.ID      Branch      City      Customer.type
Length :1000    Length :1000    Length :1000    Length :1000
N.unique :1000  N.unique : 3    N.unique : 3    N.unique : 2
N.blank : 0     N.blank : 0     N.blank : 0     N.blank : 0
Min.nchar: 11   Min.nchar: 4     Min.nchar: 6     Min.nchar: 6
Max.nchar: 11   Max.nchar: 5     Max.nchar: 9     Max.nchar: 6

Gender          Product.line  Unit.price  Quantity
Length :1000    Length :1000    Min. :10.08    Min. : 1.00
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Max.nchar: 6     Max.nchar: 22    3rd Qu.:77.94    3rd Qu.: 8.00
Max. :99.96     Max. :10.00

Tax.5.          Sales          Date          Time
Min. : 0.5085    Min. : 10.68    Length :1000    Length :1000
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Mean :15.3794     Mean : 322.97   Min.nchar: 8     Min.nchar: 10
3rd Qu.:22.4453    3rd Qu.: 471.35  Max.nchar: 9     Max.nchar: 11
Max. :49.6500     Max. :1042.65

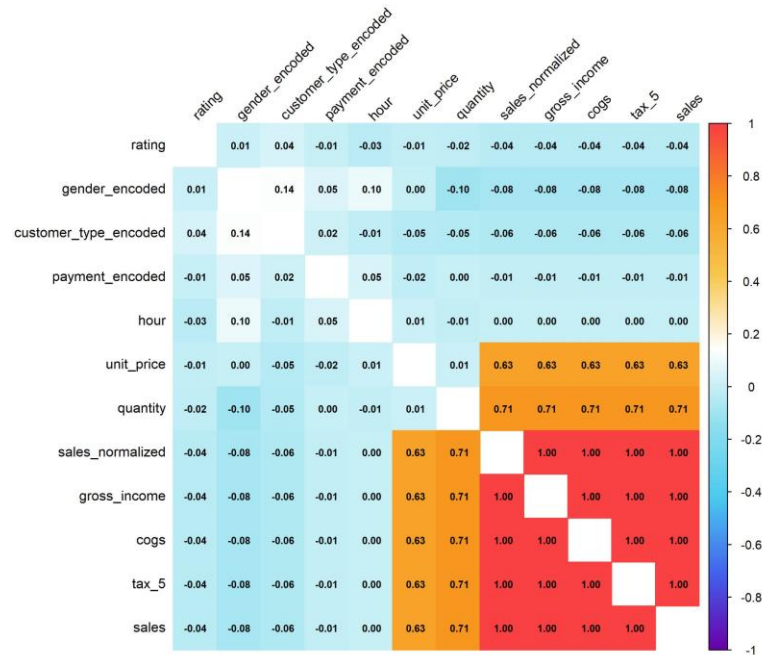
Payment         cogs          gross.margin.percentage gross.income
Length :1000    Min. :10.17    Min. :4.762     Min. : 0.5085
N.unique : 3     1st Qu.:118.50  1st Qu.:4.762    1st Qu.: 5.9249
N.blank : 0     Median :241.76  Median :4.762     Median :12.0880
Min.nchar: 4     Mean :307.59   Mean :4.762     Mean :15.3794
Max.nchar: 11    3rd Qu.:448.90  3rd Qu.:4.762    3rd Qu.:22.4453
Max. :993.00     Max. :4.762     Max. :49.6500

Rating
Min. : 4.000
1st Qu.: 5.500
Median : 7.000
Mean : 6.973
3rd Qu.: 8.500
Max. :10.000
```

Summary statistics give details about the average, spread, and distribution of numerical variables. The structure function identifies the data types for each variable.

8. Correlation Analysis

Figure 7. Correlation Heatmap



Numerical variables were analyzed using Pearson correlation analysis to explore their relationships. Variables with zero variance were deleted before computing the correlation matrix. Sales, COGS, Gross Income, and Tax had strong positive correlations. This is because of the mathematical relationship between these variables.

9. Initial Findings

- The dataset did not have any missing values.
- No duplicate records found.
- Column names and types were normalized.
- Feature engineering created new variables for month, day, hour, and time period.
- The dataset was prepared for future analysis through normalization and encoding.
- The correlation analysis showed the relationship between numerical variables.

10. Conclusion

The supermarket sales dataset has been cleaned and preprocessed using R successfully. Missing values, duplicates, and outliers were checked in the dataset before normalization, encoding, and feature engineering techniques were applied. A preliminary statistical analysis revealed that the data is consistent and suitable for advanced exploratory analysis and visualization in the next stages of the internship.

